

Question Bank

Subject: Information Security

Unit – 1 & 2

1. Consider a Hill cipher with $m=3$ with key $K = \begin{bmatrix} 25 & 3 & 7 \\ 5 & 9 & 21 \\ 11 & 8 & 13 \end{bmatrix}$
What is the ciphertext corresponding to the plaintext “VOWEL”? – 4 marks
2. Define the term information security. List and briefly define the categories of security attacks and services. – 7 marks
3. Using Vigenere cipher, encrypt the word ‘cryptography’ using the key house. – 4 marks
4. How security of polyalphabetic cipher is improved over monoalphabetic cipher? – 4 marks
5. The encryption key in a transposition cipher is (3,2,6,1,5,4). Find the decryption key. – 4 marks
6. Explain different type of cryptanalytic attack based on the amount of information known to the cryptanalyst. – 7 marks
7. Using the playfair matrix with the key “Lieutenant” encrypt the message “nest in tree”. – 3 marks
8. A generalization of the Caesar cipher, known as the affine Caesar cipher where ciphertext letter $C = E([a,b],p) = (a \cdot p + b) \bmod 26$, and $[a,b]$ is the key. A basic requirement for any encryption algorithm is that it be one-to-one. That is, if $p \neq q$, then $E(k, p) \neq E(k, q)$. The affine Caesar cipher is not one-to-one for all values of 'a'. For example, for $a = 2$ and $b = 3$, then $E([a,b], 0) = E([a, b], 13) = 3$.
Are there any limitations on the value of 'b'? Prove why or why not. – 4 marks
9. Explain one time pad (OTP). What is the drawback of OTP? – 4 marks

Unit – 3

1. Explain internal structure of single round of DES algorithm. – 7 marks
2. Describe SubBytes, ShiftRows, MixColumns, AddRoundKey in AES (Advanced Encryption standard). – 7 marks
3. What is the purpose of S-boxes in DES? Explain the avalanche effect. – 3 marks
4. Compare the substitution in AES and DES. Why do we have only one substitution table(S-box) in AES, but several in DES? – 3 marks
5. Explain the byte substitution in AES (Advanced Encryption Standard). – 4 marks
6. What is diffusion and confusion? Explain Feistel encryption and decryption. – 7 marks
7. Compare the permutations in DES and AES. Why expansion and compression permutations are needed in DES, but not in AES? – 3 marks
8. How diffusion and confusion is achieved in DES (Data Encryption Standard)? Explain single round of DES algorithm. – 7 marks
9. Differentiate between block cipher and stream cipher. List out and briefly explain parameters and features on which Feistel structure is dependent – 7 marks
10. Differentiate between differential and linear cryptanalysis. – 4 marks
11. Explain inputs for single AES round. – 7 marks
12. Explain AES key expansion. – 7 marks
13. Explain meet-in the-middle attack. Explain Triple DES with two keys. – 7 marks
14. Explain Electronic codebook mode (ECB) and Cipher block chaining mode (CBC). – 7 marks

15. Explain cipher feedback mode (CFB) and output feedback mode (OFB). – 7 marks
16. Explain Cipher feedback mode (CFB) and counter mode. – 7 marks

Unit – 4 & 5

1. Explain public key cryptosystem and compare conventional encryption with public key encryption. – 7 marks
2. What are the requirements for public key cryptography? Explain cryptanalysis for public-key. – 7 marks
3. Explain the following two issues of the complexity of the computation required to use RSA. – 7 marks
 - i) Encryption/decryption
 - ii) Key generation.
4. RSA algorithm is vulnerable to which cryptanalytic attack? Justify your answer. – 3 marks
5. Write the possible approaches to attack RSA. – 7 marks
6. Two prime numbers for RSA are 11 and 17. Value of e is 7. Find the public key and private key using RSA algorithm. – 4 marks
7. Explain Diffie Hellman key exchange scheme in detail. – 4 marks
8. Explain man-in-the-middle attack for Diffie Hellman key exchange. – 4 marks

Unit – 6

Hash Function

1. Write the three desirable properties of hash functions. – 4 marks
2. Write the properties needed for a hash function H to be useful for message authentication. Is hash function resistant against birthday attack? – 7 marks
3. How can we use block ciphers as Hash functions? – 4 marks
4. What is the difference between weak and strong collision resistance? Explain the general principle used in all hash functions. – 7 marks
5. Explain Message digest generation using SHA (Secure Hash algorithm). – 7 marks
6. Illustrate various ways in which a hash code can be used to provide message authentication. – 7 marks

MAC

1. What types of attacks are addressed by message authentication? – 4 marks
2. What is message authentication code? Describe the situations in which message authentication code is used. – 7 marks
3. What are some approaches to produce message authentication? – 7 marks
4. Write MAC (Message Authentication code) requirements. – 4 marks

Digital Signature

1. What is the difference between direct and arbitrated digital signature? – 3 marks
2. What are the properties digital signatures should have? Write digital signature algorithm. – 7 marks
3. What are some threats associated with a direct digital signature scheme? – 4 marks

Unit – 7

1. What is the difference between link and end-to-end encryption? – 3 marks
2. Explain the key distribution scenario in which each user shares a unique master key with key distribution centre. – 4 marks
3. What is the difference between a session key and a master key? – 3 marks
4. Explain decentralized key distribution – 3 marks
5. What are the different methods for distributing public keys? – 7 marks
6. Explain symmetric key distribution using asymmetric encryption. – 7 marks
7. Discuss about lifetime of a session key – 3 marks
8. List out and explain PKIX management functions. – 7 marks
9. Explain X.509 certificates formats. – 7 marks

Unit – 8

1. Explain different types of replay attacks. What are the different methods to cop up with replay attacks? – 7 marks
2. What is suppress-replay attack? – 3 marks
3. What is the difference between mutual authentication and one-way authentication? – 4 marks
4. Explain remote user authentication technique using Kerberos. – 7 marks
5. Explain remote user authentication technique using Asymmetric encryption. – 7 marks
6. What is the role of ticket granting server and authentication server in Kerberos? – 4 marks